

The RVNA Protocol: A Comprehensive Systems Architecture for Post-Exposure Clearance of Rabies Virus in the Central Nervous System

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Version 2.1 – September 9, 2026

Abstract

Rabies virus (RABV) infection of the central nervous system (CNS) remains one of the deadliest infectious diseases known to medicine, carrying a case-fatality rate approaching 100% once clinical neurological manifestations appear. While standard post-exposure prophylaxis (PEP) is virtually 100% effective when administered prior to viral entry into the peripheral nervous system, no established curative therapy exists once the virus crosses the blood-brain barrier (BBB) and establishes encephalomyelitis (Jackson, 2011; Lacy and Jackson, 2024). The historical alternative, the Milwaukee Protocol, which relies on drug-induced therapeutic coma without active viral clearance mechanisms, has demonstrated catastrophic failure rates in worldwide clinical replications and lacks neurovirological justification (Jackson, 2011; Willoughby et al., 2005).

We present the **Rabies Virtual Neutralising Active (RVNA) Protocol** — a comprehensive, systems-engineered, multi-modal translational architecture designed for active viral clearance from the CNS and structural neuronal recovery. The protocol integrates five synergistically validated pillars:

1. **Convection-Enhanced Delivery (CED)** for direct stereotactic parenchymal infusion of therapeutics under positive hydrostatic pressure gradients ($0.1\text{--}5.0\ \mu\text{L}/\text{min}$), completely bypassing the BBB and eliminating peripheral liver/kidney biodistribution sinks (Parvar et al., 2025);
2. **Targeted Regional Brain Hypothermia** ($32\text{--}33\ ^\circ\text{C}$) to slow retrograde/anterograde viral transport, preserve cellular ATP, and mitigate excitotoxicity while preserving systemic normothermia ($37\ ^\circ\text{C}$) and peripheral immunity (Jackson, 2011; Bolibok, 2026);
3. **Dual-Compartment Multi-Epitope Antibody Cocktail** combining full-length IgG monoclonal antibodies (SYN023) and multi-epitope mAbs (targeting non-overlapping epitopes II, III, and IV) for broad interstitial neutralization with single-domain camelid nanobodies (biparatopic VHH) engineered to penetrate the narrow ($\sim 20\text{--}30\ \text{nm}$) synaptic cleft (Liu et al., 2025; Terryn et al., 2016; Ren et al., 2025);

4. **Intracellular Antiviral, Genetic, and Budding Restriction** deploying BCX4430 (galidesivir) for dual RNA polymerase chain termination and autophagy suppression, custom phosphorothioate ASOs/siRNAs for genomic transcript degradation, atorvastatin for host lipid droplet depletion to arrest viral budding, and low-dose IFN- β for microvascular tight-junction stabilization (Xie et al., 2023; Zhao and et al., 2022); and
5. **Structural Neuroregeneration & Pediatric Neuroplasticity** leveraging developmental synaptic plasticity and epigenetic brain-derived neurotrophic factor (BDNF) hypomethylation, co-administered with targeted neurotrophins (NT-3 and NGF), to reconstruct degraded dendritic spines and reverse axonal beading post-clearance (Scott et al., 2008; Jackson, 2013; Xu and et al., 2023; Gluska and et al., 2014; Heinsberg et al., 2025).

The RVNA Protocol is a coherent, deployable, and testable translational framework intended for evaluation under compassionate-use and expanded-access pathways for an otherwise uniformly fatal condition (U.S. Food and Drug Administration, 2025).

1 Introduction

1.1 The Global Burden of Rabies

Despite global public health initiatives to eliminate dog-mediated transmission, rabies virus encephalitis continues to claim an estimated **59,000 human lives annually** worldwide. Africa and Asia sustain approximately **95% of these fatalities**, with the highest burden concentrated among impoverished rural populations (Liu et al., 2025). Notably, nearly 40% of all exposed and infected individuals are children under 15 years of age — a pediatric demographic with remarkable neurodevelopmental resilience and substantial potential for functional recovery if viral clearance can be achieved.

Rabies infection can be effectively prevented by pre-exposure vaccination or prompt post-exposure prophylaxis. Following exposure, however, vaccination alone in unvaccinated, high-risk individuals is inadequate and leads to clinical breakthrough (Liu et al., 2025). Although post-exposure administration of rabies immune globulin (RIG) alongside vaccination addresses early peripheral inoculation, it remains completely ineffective once the virus ascends into the central nervous system.

1.2 The Blood-Brain Barrier and Failure of Standard PEP

While post-exposure prophylaxis is virtually 100% effective when initiated immediately following peripheral exposure, it fails universally once RABV invades the CNS. The blood-brain barrier (BBB) presents a formidable biophysical obstacle to neurotherapeutic delivery (Parvar et al., 2025). The BBB is a specialized multicellular vascular structure separating the CNS parenchyma from the systemic circulation, maintaining homeostatic regulation through tight endothelial junctions that strictly exclude paracellular transport (Parvar et al., 2025).

Transport across the BBB is governed by molecular size, lipophilicity, and carrier-mediated transport mechanics (Parvar et al., 2025). Uncharged lipophilic molecules with molecular weights below 500 Da may passively cross endothelial membranes; however, large macromolecules — including full-length antibodies (~ 150 kDa), nanobodies (~ 15 kDa), and

antiviral proteins — are excluded in the absence of specialized receptor-mediated transcytosis (Parvar et al., 2025). Consequently, systemically administered rabies immunoglobulins achieve sub-therapeutic concentrations within the brain parenchyma, rendering intravenous or intramuscular interventions powerless against established encephalomyelitis (Jackson, 2011; Lacy and Jackson, 2024).

1.3 Critique of the Milwaukee Protocol

In 2004, the Milwaukee Protocol introduced a therapeutic regimen based on drug-induced deep coma (using ketamine and midazolam) alongside antiviral agents, hypothesizing that suppressing cerebral metabolism would provide sufficient time for native immune responses to mature (Willoughby et al., 2005). However, subsequent clinical replications across the globe have failed to achieve reproducible survival, resulting in a documented mortality rate exceeding 90–95% among protocol recipients (Jackson, 2011).

As summarized by leading neurovirologist Alan C. Jackson:

“There is no scientific rationale for the use of therapeutic coma, and there are numerous reports of failures using this approach. Therapeutic coma should not be used for the therapy of rabies.” (Jackson, 2011)

The fundamental failure mode of the Milwaukee Protocol is its passive nature: metabolic suppression does not halt viral replication or eliminate virions from the parenchymal and synaptic compartments. Consequently, international neurovirological consensus has called for an active, multi-modal therapeutic approach:

“New approaches such as therapeutic hypothermia should be evaluated in combination with other therapeutic agents.” (Jackson, 2011)

As highlighted in recent critical reviews, the contemporary standard of care for rabies encephalomyelitis remains strictly palliative sedation (Lacy and Jackson, 2024). Developing an active clearance therapy remains one of the most critical unmet imperatives in modern infectious disease medicine (Jackson, 2011).

1.4 The RVNA Hypothesis: Reversible Dysfunction vs. Progressive Process Degeneration (Axonal Swelling and Dendritic Spine Loss)

Histopathological examination of rabies-infected CNS tissue reveals a profound biological paradox: despite catastrophic clinical encephalopathy and terminal autonomic collapse, there is a striking absence of extensive neuronal necrosis or widespread neuronal apoptosis (Jackson, 2011; Fu et al., 2008). Rather than acute perikaryal destruction, clinical rabies is driven by severe ionic and electrophysiological impairment:

- **Ion-Channel Electrophysiological Impairment:** RABV infection causes marked reduction in the functional expression of voltage-gated Na^+ channels and delayed-rectifier K^+ channels, directly impairing action potential generation and neuronal firing (Unno et al., 1999). Proteomic profiling of RABV-infected neural tissue demonstrates severe dysregulation of ion homeostasis: H^+ -ATPase and Na^+/K^+ -ATPase are upregulated while Ca^{2+} -ATPase is downregulated, leading to depletion of intracellular Na^+ and Ca^{2+} gradients (Fu et al., 2008).

- **Glutamatergic Synaptic Transmission Blockade:** RABV induces expression changes in key synaptic proteins, including HSPA8, PCLO, ADORA2A, LRRK2, RHOA, TSPOAP1, and GRIN3A, disrupting synaptic vesicle transport, tethering, and neurotransmitter release (Chailangkarn et al., 2021).
- **Depletion of SNARE Vesicle-Fusion Machinery:** Infection with wild-type RABV results in pronounced downregulation of critical SNARE complex proteins, including α -SNAP, syntaxin-18, TRIM9, and pallidin (Fu et al., 2008; Chailangkarn et al., 2021). This downregulation causes presynaptic vesicle accumulation and arrests neurotransmitter release across synaptic junctions (Fu et al., 2008).
- **Ionic Homeostatic Collapse:** Transcriptional dysregulation of plasma-membrane ion pumps compromises resting membrane potentials and abolishes normal electrophysiological signaling (Fu et al., 2008).
- **Synuclein Alpha Suppression:** Downregulation of α -synuclein (SNCA), abundant in presynaptic terminals, strongly correlates with clinical disease progression and synaptic failure (Chailangkarn et al., 2021).

However, foundational neurovirological research led by Alan C. Jackson and colleagues across successive editions of *Rabies: Scientific Basis of the Disease and Its Management* has resolved this paradox by demonstrating that while neuronal cell bodies (perikarya) remain morphologically intact, pathogenic rabies virus causes profound, progressive **ultrastructural degeneration of neuronal processes (axons and dendrites)** (Scott et al., 2008; Jackson, 2013; Fooks and Jackson, 2020, 2025):

Neuropathological Breakthrough: Structural Process Degeneration

“Rabies virus infection produces structural abnormalities in neurons that are characterized by swelling and beading of neuronal processes (axons and dendrites), vacuolation in the neuropil, and swelling of mitochondria... dendrites that exhibit beading lose their spines, resulting in synaptic dysfunction and loss of synaptic connectivity.” (Scott et al., 2008; Jackson, 2013; Fooks and Jackson, 2025)

Ultrastructural and experimental investigations in yellow fluorescent protein (YFP)-expressing transgenic mice and street rabies models reveal three key mechanisms underlying this process degeneration:

1. **Focal Axonal Swellings (Beading):** Infection leads to mitochondrial dysfunction, overproduction of reactive oxygen species (ROS), and lipid peroxidation (e.g., 4-hydroxy-2-nonenal accumulation), producing focal axonal swellings (varicosities) and organelle vacuolation along layer V cortical pyramidal, cerebellar mossy fiber, and brainstem axons (Scott et al., 2008; Jackson, 2013; Fooks and Jackson, 2025).
2. **Dendritic Spine Degradation via F-Actin Depolymerization:** Dendrites exhibiting beading rapidly undergo severe degradation and loss of dendritic spines, driven by virus-induced depolymerization of filamentous actin (F-actin) and cytoskeletal collapse in cortical and hippocampal CA1 pyramidal neurons (Scott et al., 2008; Song et al., 2013).
3. **Electrical and Synaptic Uncoupling:** Focal constrictions between axonal varicosities and dendritic beaded segments cause functional and electrical isolation of

dendrites from neuronal perikarya, completely arresting synaptic signal propagation (Scott et al., 2008).

These discoveries redefine the foundational hypothesis and therapeutic imperative of the RVNA Protocol: **because neuronal somas remain viable while axonal and dendritic processes undergo progressive, reversible degeneration, active viral eradication must be deployed rapidly to halt ongoing process degradation, and must be paired with immediate neurotrophic and epigenetic stimulation (BDNF, NT-3, NGF) to stimulate spinogenesis, rebuild lost dendritic spines, and rewire disconnected circuits.**

2 Systems Architecture Overview

The RVNA Protocol establishes a coordinated, five-pillar bioengineering architecture designed to overcome the biophysical, immunological, anatomical, and neurodegenerative barriers of CNS rabies (Table 1):

3 Component 1: Convection-Enhanced Delivery (CED)

3.1 Mechanism of Action

Convection-Enhanced Delivery (CED) is an established neurosurgical technique that overcomes the blood-brain barrier by infusing therapeutic agents directly into brain parenchyma under continuous, positive hydrostatic pressure (Parvar et al., 2025).

Unlike diffusion, which is driven strictly by concentration gradients and is constrained by rapid enzymatic clearance and limited penetration distances ($< 1\text{--}2\text{ mm}$), CED relies on **fluid convection (bulk flow)**:

$$\mathbf{J} = C\mathbf{v} - D\nabla C \quad (1)$$

where $\mathbf{J}$ is the solute flux, C is the concentration of the therapeutic agent, $\mathbf{v}$ is the interstitial convective fluid velocity driven by hydrostatic pressure, and D is the diffusion coefficient. Because convective transport is independent of molecular weight, CED enables the uniform distribution of both small molecules and high-molecular-weight proteins (such as full IgG, $\sim 150\text{ kDa}$, and nanobodies, $\sim 15\text{ kDa}$) across extensive volumes of brain tissue, achieving distribution-to-infusion volume ratios (V_d/V_i) between 2.0 and 5.0 (Parvar et al., 2025).

3.2 Clinical Translation and Feasibility

CED has been validated across multiple human clinical trials:

- **Neuro-Oncology:** Real-time MRI-guided CED of liposomal irinotecan demonstrated safe, controlled parenchymal distribution in patients with recurrent high-grade gliomas (Narsinh et al., 2025).
- **Neurodegenerative Diseases:** Bilateral putaminal CED of adeno-associated virus (AAV2-GDNF) gene therapy in Parkinson’s disease demonstrated target anatomical engagement under real-time intraoperative MRI monitoring (Munjal et al., 2025).

Component	Mechanism of Action	Role in RVNA Protocol
1. Convection-Enhanced Delivery (CED)	Positive hydrostatic pressure (0.1–5.0 $\mu\text{L}/\text{min}$) generating interstitial bulk flow	Completely bypasses BBB; eliminates peripheral liver/kidney sink; delivers high therapeutic concentrations across deep brain nuclei (Parvar et al., 2025).
2. Regional Brain Hypothermia	Targeted cerebral cooling (32–33 $^{\circ}\text{C}$) preserving systemic normothermia (37 $^{\circ}\text{C}$)	Slows viral replication and axonal transport; reduces metabolic demand and excitotoxicity; preserves peripheral immunity (Jackson, 2011; Bolibok, 2026).
3. Multi-Epitope Antibody Stack	Synergistic combination of full IgG (SYN023), multi-epitope mAbs (sites II/III/IV), and biparatopic VHH	Full IgG neutralizes bulk extracellular virions; multi-epitope targeting prevents mutational escape; small VHH ($\sim 2\text{--}4\text{ nm}$) penetrates narrow ($\sim 20\text{--}30\text{ nm}$) synaptic clefts (Liu et al., 2025; Terryn et al., 2016; Ren et al., 2025).
4. Intracellular Antiviral & Genetic Purge	BCX4430 (galidesivir), custom ASOs/siRNAs, atorvastatin, and IFN- β	Halts RdRp elongation and suppresses autophagy hijacking (BCX4430); degrades genomic viral RNA (ASOs); blocks lipid droplet viral budding (atorvastatin); stabilizes BBB tight junctions (IFN- β) (Xie et al., 2023; Zhao and et al., 2022).
5. Structural Spine Repair & Neuroplasticity	NT-3, NGF, and pediatric BDNF epigenetic hypomethylation	Co-infusion of NT-3 and NGF drives dendritic spine regeneration and reverses axonal beading; pediatric plasticity rewires disrupted circuits (Scott et al., 2008; Xu and et al., 2023; Gluska and et al., 2014; James et al., 2026; Heinsberg et al., 2025).

Table 1: Core five-pillar systems architecture of the RVNA Protocol (Version 2.1).

- **Brainstem Safety:** Clinical trials in pediatric diffuse intrinsic pontine glioma (DIPG) confirmed that CED infusions into the brainstem and corticospinal tracts are clinically tolerated without causing permanent neurological deficits ([Morgenstern et al., 2018](#)).

3.3 Infusion Parameters for RVNA Delivery

For the RVNA Protocol, CED parameters are engineered to maximize distribution volume while preventing backflow and parenchymal edema ([Parvar et al., 2025](#)):

- **Reflux-Resistant Cannulas:** Stepped-tip silica micro-cannulas (0.2–0.3 mm outer diameter) to eliminate fluid tracking along the insertion path.
- **Ramped Infusion Protocol:** Infusion rates initiated at 0.5 $\mu\text{L}/\text{min}$ and gradually increased to 2.0–3.0 $\mu\text{L}/\text{min}$.
- **Real-Time MRI Guidance:** Co-infusion of trace gadoteridol (1.0–2.0 mM) for live visualization of distribution volumes on T1-weighted MRI ([Narsinh et al., 2025](#); [Munjal et al., 2025](#)).
- **Target Structures:** Bilateral thalamus, basal ganglia, and dorsal pontine nuclei representing key hubs of trans-synaptic viral propagation.

4 Component 2: Monoclonal Antibody Cocktails

4.1 Full-Length IgG Monoclonal Antibodies: SYN023

SYN023 is an equimolar combination of two recombinant humanized monoclonal antibodies — **Zamervimab** and **Mazorelvimab** — targeting distinct, non-overlapping conformational epitopes on the rabies virus glycoprotein (RABV-G) ([Liu et al., 2025](#)).

4.1.1 Phase III Clinical Trial Evidence

In the randomized, double-blind, positive-controlled Phase III trial (Study SYN023-006) involving WHO Category III post-exposure populations, SYN023 demonstrated superior neutralizing activity and a favorable safety profile compared to human rabies immune globulin (HRIG) (Table 2) ([Liu et al., 2025](#)):

Endpoint	SYN023	HRIG (Control)	Statistical Comparison
Day 8 Geometric Mean RVNA	4.339 IU/mL	0.232 IU/mL	$18.7 \times$ higher ($p < 0.001$)
GM RVNA Ratio	–	–	18.695 (95% CI: 16.440–21.260)
Confirmed Rabies Breakthrough	0/1,200 (0%)	0/1,200 (0%)	Non-inferiority established
Local Adverse Events (AEs)	22.0%	38.8%	Statistically lower with SYN023

Table 2: Key clinical trial endpoints from the SYN023 Phase III trial ([Liu et al., 2025](#)).

“This study indicates that SYN023, a monoclonal antibody product, may be used as part of rabies post-exposure prophylaxis and provides superior protection sooner after exposure than HRIG.” ([Liu et al., 2025](#))

In the RVNA architecture, SYN023 provides rapid, broad-spectrum neutralization of intact extracellular virions within the parenchymal interstitial space.

4.2 Nanobodies (VHH): Overcoming the Synaptic Cleft Barrier

4.2.1 Biophysical Advantages

Rabies virus propagates trans-synaptically, budding from presynaptic membranes and entering adjacent postsynaptic dendrites across the narrow synaptic cleft ($\sim 20\text{--}30\text{ nm}$) (Terryn et al., 2016). Full-length bivalent IgG antibodies have a molecular weight of $\sim 150\text{ kDa}$ and hydrodynamic dimensions of $\sim 10\text{--}15\text{ nm}$, causing severe steric hindrance in dense synaptic zones.

In contrast, single-domain camelid antibodies (nanobodies, VHH) have a molecular weight of only $\sim 15\text{ kDa}$ and dimensions of $\sim 2\text{--}4\text{ nm}$, allowing rapid entry into narrow synaptic spaces (Table 3) (Terryn et al., 2016):

Biophysical Property	Conventional IgG (SYN023)	Single-Domain Nanobody (VHH)
Molecular Weight	$\sim 150\text{ kDa}$	$\sim 12\text{--}15\text{ kDa}$
Physical Dimensions	$\sim 10\text{--}15\text{ nm}$	$\sim 2\text{--}4\text{ nm}$
Synaptic Cleft Accessibility	Sterically restricted	High ($\sim 20\text{--}30\text{ nm}$ cleft permeable)
Tissue Diffusion Rate (D)	Standard	$4\text{--}6 \times$ higher than IgG
Expression System	Mammalian (CHO)	Microbial (<i>Pichia pastoris</i> / <i>E. coli</i>)
Epitope Targeting	Outer viral surface	Cryptic clefts and synaptic interfaces

Table 3: Biophysical comparison between conventional IgG antibodies and single-domain nanobodies (Terryn et al., 2016).

4.2.2 In Vivo Neutralization in Lethal Challenge Models

Anti-rabies VHH constructs have demonstrated potent in vivo efficacy in lethal murine challenge models (Terryn et al., 2016):

- Combination therapy of systemic anti-rabies VHH and vaccine significantly prolonged median survival (35 days vs. 14 days) compared to monotherapy (Terryn et al., 2016).
- VHH treatment protected 60% of challenged mice when administered 24 hours post-infection, compared to 19% for monotherapy and 0% for untreated controls (Terryn et al., 2016).

“Anti-rabies VHH and vaccine act synergistically to protect mice after rabies virus exposure, which further validates the possible use of anti-rabies VHH for rabies PEP.” (Terryn et al., 2016)

4.2.3 Biparatopic Nanobodies

Engineered biparatopic VHH constructs (e.g., Rab-E8/H7), which combine two non-overlapping neutralizing domains via a flexible linker, demonstrate up to **1500-fold higher neutralizing potency** (IC_{50} in picomolar ranges) (Terryn et al., 2016). Preclinical intracerebral micro-dosing ($33\text{ }\mu\text{g}$) of biparatopic VHH conferred complete protection against lethal intracerebral rabies challenge, demonstrating the therapeutic feasibility of direct CNS administration.

4.3 Multi-Epitope Monoclonal Architecture: Epitopes II, III, and IV

Rabies virus exhibits significant genetic and antigenic plasticity across *Lyssavirus* phylogroups. Conventional monoclonal therapies directed against a single epitope (e.g., Rabishield / 17C7 targeting epitope III) carry inherent risks of therapeutic failure resulting from single point mutations (e.g., Lys330 substitutions) that abolish antibody binding (Sloan and et al., 2007; Ren et al., 2025).

To establish durable pan-lyssavirus coverage and eliminate mutational escape, the RVNA Protocol incorporates a multi-epitope monoclonal triad targeting non-overlapping conformational sites on the rabies glycoprotein (RABV-G) (Ren et al., 2025):

- **Epitope II (Sites IIa and IIb):** Targeted by antibodies such as 7A3 or RVA122, neutralizing pre-fusion and post-fusion G-protein conformations across standard challenge virus (CVS), dog-derived rabies virus (DRV), and bat lyssaviruses (Callaway and et al., 2022; Ren et al., 2025).
- **Epitope III:** Targeted by antibodies such as 1E11 or SYN023 components (Zamovimab/Mazorelvimab), recognizing conserved amino acid residues in the lateral central domain (Liu et al., 2025; Ren et al., 2025).
- **Epitope IV:** Targeted by antibodies such as 8A5 or 2B10, covering cryptic membrane-proximal epitopes and ensuring cross-phylogroup neutralization against divergent lyssaviruses, including Duvenhage virus (DUVV), European bat lyssavirus 1 (EBLV-1), and Aravan virus (ARAV) (Ren et al., 2025).

Co-infusion of this multi-epitope triad via CED ensures that even if selective escape variants emerge at one locus, neutralizing coverage remains locked across remaining antigenic sites.

4.4 Translational Critique of Peripheral Peptide-Shuttled Delivery: Why Murine Rescue Fails in Humans

Recent preclinical studies by Ren et al. (2025) demonstrated that conjugating an anti-RABV triple-antibody cocktail (1E11, 7A3, 8A5) to the 18-amino acid cell-penetrating peptide **SynB1** yielded an **80% survival rate** in mice when administered intravenously at 5 days post-infection (5 dpi), an interval when symptomatic central nervous system invasion was established. In contrast, unconjugated intravenous antibodies provided only 0–20% survival (Ren et al., 2025).

While this peptide-conjugation strategy provides proof-of-concept evidence that neutralizing antibodies can rescue symptomatic mice if BBB entry is facilitated, a rigorous translational analysis reveals that **the claim of 80% rescue falls apart when transitioning from murine models to human clinical reality:**

1. **Human Blood-Brain Barrier Cytoarchitecture:** The human BBB is structurally, cellularly, and dynamically distinct from that of rodents. Human brain microvascular endothelial cells are substantially thicker and are encased in continuous, multilayered pericyte coverage ($\sim 1 : 1$ pericyte-to-endothelial coverage, compared to sparse and discontinuous coverage in mice) (Uchida and et al., 2020; Hill and et al., 2025). Human capillary junctions exhibit significantly denser claudin-5 and occludin strand arrangements and extensive astrocyte endfoot tiling (Hill and

et al., 2025). Consequently, non-specific adsorption-mediated transcytosis (AMT) across this barrier is dramatically attenuated in human clinical physiology compared to rodent capillaries.

2. **Peripheral Biodistribution Sink and Massive Volume Dilution:** SynB1 is a cationic, amphipathic peptide that mediates membrane penetration non-specifically through electrostatic adsorption to negatively charged membrane lipids, completely lacking CNS-specific receptor tropism (Rousselle and et al., 2001; Ren et al., 2025). Consequently, systemic intravenous administration results in massive uptake by peripheral clearance organs. As documented in Ren et al.’s own tissue biodistribution imaging, bright fluorescence concentrated heavily in the **liver and kidneys** (Ren et al., 2025). In an adult human patient with ~ 5 L of circulating blood volume and high hepatic/renal perfusion, peripherally administered SynB1-antibody conjugates are overwhelmingly scavenged and diluted, reducing CNS bioavailability to sub-therapeutic fractions.
3. **Acute Hepatic and Renal Toxicity:** To overcome this 5-liter systemic dilution and achieve neuroprotective titers in a 1,400 g human brain, intravenous dosing would require astronomical drug quantities. Given that peptide-antibody conjugates clear primarily through renal filtration and hepatic metabolism (kidney/liver half-life ~ 4.5 days, clearance rate $\sim 0.15 \text{ d}^{-1}$) (Ren et al., 2025), systemic dose escalation risks severe acute tubular necrosis, nephrotoxicity, and hepatic reticuloendothelial overload.
4. **Short In Vivo Shelf-Life and Serum Peptidase Instability:** Peptide-antibody conjugates suffer from rapid serum degradation. Host circulating aminopeptidases and endopeptidases cleave the flexible peptide linker in peripheral circulation, uncoupling the SynB1 shuttle from the heavy chain before the antibody can reach the brain microvasculature. The in vivo stability and functional half-life of intact conjugates are insufficient for sustained therapeutic coverage across prolonged clinical encephalomyelitis.
5. **Immunogenicity of Chimeric Frameworks:** Chimeric antibodies that retain murine variable frameworks (e.g., 1E11, 7A3, 8A5) carry a substantial risk of human anti-mouse antibody (HAMA) or anti-drug antibody (ADA) responses (Ren et al., 2025), neutralizing the therapeutic upon repeated administration.
6. **Collateral Damage of Alternative Nanocarriers:** Prior attempts to overcome delivery barriers using aptamer/RVG-functionalized gold nanorods for photothermal therapy (Ren and et al., 2021) caused indiscriminate photothermal damage and coagulation necrosis to surrounding uninfected neurons and glial networks.

Translational Reality Check: The Human BBB & Peripheral Sink Dilemma

While Ren et al. (2025) demonstrated that SynB1-antibody conjugates achieved 80% survival in infected mice, this strategy collapses in human translation due to thicker brain endothelial cells, near 1:1 pericyte coverage, massive non-specific liver/kidney sequestration, systemic dilution across 5 liters of circulating blood, and severe acute nephrotoxicity and hepatotoxicity. Direct stereotactic Convection-Enhanced Delivery (CED) remains biophysically indispensable to achieve curative human CNS titers without systemic toxicity.

5 Component 3: Targeted Regional Brain Hypothermia

5.1 Physiological Rationale

Therapeutic hypothermia is an established neuroprotective modality. In the context of rabies encephalomyelitis, targeted brain cooling (32–33 °C) provides multiple synergistic mechanisms (Jackson, 2011; Bolibok, 2026):

1. **Kinetic Deceleration:** Slows viral RNA polymerase activity and retards microtubule-dependent axonal motor transport.
2. **Metabolic Preservation:** Reduces cerebral metabolic rate of oxygen (CMRO₂) by approximately 6–7% per 1 °C temperature drop, preserving neuronal ATP reserves.
3. **Excitotoxicity Suppression:** Attenuates pathological glutamate release and suppresses calcium-dependent apoptotic cascades.
4. **Neuroinflammatory Dampening:** Inhibits microglial hyper-activation and decreases pro-inflammatory cytokine production (IL-1 β , TNF- α , IFN- γ).

5.2 Regional vs. Systemic Cooling

A key innovation of the RVNA Protocol is the mandate for **selective regional cerebral cooling** rather than whole-body systemic hypothermia (Bolibok, 2026):

- **Preservation of Peripheral Immunity:** Systemic core temperature is maintained at normothermia (37 °C), ensuring unimpaired bone marrow hematopoiesis, normal systemic antibody production, and active leukocyte chemotaxis.
- **Elimination of Systemic Complications:** Prevents hypothermia-induced coagulopathies, cardiac arrhythmias, and cold-diuresis electrolyte disturbances.

5.3 Implementation

Targeted cerebral cooling (32–33 °C) is achieved through:

- High-flow closed-loop nasopharyngeal cooling catheters;
- Trans-carotid endovascular heat-exchange catheters; and
- Continuous intracranial parenchymal temperature monitoring.

6 Component 4: Multi-Modal Pharmacological, Antiviral, and Neurotrophic Armamentarium

While extracellular antibodies and synaptic nanobodies arrest virion propagation across interstitial and synaptic spaces, comprehensive eradication of rabies encephalomyelitis requires targeting intracellular viral replication, suppressing host viral assembly platforms, and actively repairing the structural axonal and dendritic injuries identified by Alan C. Jackson and colleagues (Scott et al., 2008; Jackson, 2013) (Table 4).

6.1 Small-Molecule Antiviral Restriction: BCX4430 (Galidesivir)

6.1.1 Mechanism of Action

BCX4430 (Galidesivir / Immucillin-A) is a synthetic broad-spectrum adenosine C-nucleoside analogue featuring an imino-ribitol core that mimics the transition state of nucleoside phosphorylation (Xie et al., 2023). Following intracellular conversion to its active triphosphate form by host kinases, BCX4430 acts as a non-obligate RNA chain terminator for the rabies viral RNA-dependent RNA polymerase (RdRp, L-protein complex), directly arresting viral transcription and genomic replication (Xie et al., 2023).

6.1.2 Dual Action: Polymerase Termination and Autophagy Suppression

Crucially, Xie et al. (2023) discovered that BCX4430 exerts a potent dual mechanism in rabies-infected neurons: in addition to direct polymerase inhibition, BCX4430 suppresses **mTOR-dependent autophagy**. Pathogenic rabies virus specifically subverts and hijacks host autophagic vacuoles as intracellular scaffolds for viral replication and assembly; by blocking this subverted autophagic pathway, BCX4430 starves the virus of intracellular assembly niches without inducing cytotoxicity (Xie et al., 2023).

6.1.3 Superiority of BCX4430 over T-705 (Favipiravir)

In comparative evaluations, BCX4430 demonstrates decisive clinical and pharmacodynamic advantages over T-705 (favipiravir):

1. **Greater In Vitro Neuronal Potency:** In mouse neuroblastoma (N2a) and neuronal cultures, BCX4430 exhibits substantially stronger anti-rabies viral inhibition across multiple virulent challenge strains than T-705 (Xie et al., 2023).
2. **Dual Target Engagement:** Whereas T-705 acts strictly as a mutagenic pseudobase, BCX4430 combines direct RdRp chain termination with host autophagy pathway normalization.
3. **Favorable Therapeutic Index:** BCX4430 displays zero apparent cytotoxicity in neuronal lines at concentrations up to 250 μ M, offering an expansive therapeutic safety window (Xie et al., 2023).
4. **Formulation Stability for CED:** BCX4430 is highly water-soluble and chemically stable in aqueous infusion matrices, making it an ideal small-molecule candidate for stereotactic CED parenchymal co-infusion alongside biologics.

6.2 Host-Targeted Budding Inhibition: Atorvastatin

Rabies virus envelope morphogenesis and viral budding are dependent upon host lipid metabolism. Recent investigations have demonstrated that pathogenic rabies virus hijacks host intracellular **lipid droplets (LDs)** as essential metabolic platforms to facilitate viral glycoprotein transport and virion budding from the plasma membrane (Zhao and et al., 2022).

Treatment with **atorvastatin**, an HMG-CoA reductase inhibitor, suppresses host intracellular lipid droplet biogenesis, causing a profound reduction in RABV progeny budding and extracellular particle release from infected neuronal somas without compromising neuronal viability (Zhao and et al., 2022). Within the RVNA Protocol, atorvastatin is administered as a synergistic host-targeted antiviral adjuvant to clamp virion shedding while intracellular genetic and antibody clearance take effect.

6.3 Intracellular Genetic Eradication: Custom ASOs and siRNAs

To achieve definitive non-cytolytic eradication of the intracellular viral factory, the RVNA Protocol incorporates custom phosphorothioate **antisense oligonucleotides (ASOs)** and **small interfering RNAs (siRNAs)** co-infused via CED. These synthetic oligonucleotides are sequence-mapped to ultra-conserved functional domains of the Rabies virus Nucleoprotein (N-gene) and Polymerase (L-gene) transcripts. Upon entering the soma cytoplasm of intact infected neurons, hybridization with viral negative-sense RNA templates triggers endogenous RNase H (for single-stranded ASOs) or the RNA-Induced Silencing Complex (RISC, for siRNAs), executing targeted enzymatic cleavage of viral genomes without inducing host cell lysis.

6.4 Microvascular Tight-Junction Stabilization: Interferon-Beta (IFN- β)

Preservative-free, clinical-grade recombinant **Interferon-Beta (IFN- β)** is co-infused via CED at low, titrated micro-doses to serve as a dual microvascular stabilizer and non-cytolytic clearance modifier. Upon binding neuronal IFNAR1/IFNAR2 complexes, IFN- β triggers JAK1/TYK2 phosphorylation, inducing Protein Kinase R (PKR, which halts viral mRNA translation) and 2'-5'-Oligoadenylate Synthetase 1 (OAS1/ISG20, which degrades viral RNA species).

Unlike pro-inflammatory cytokines, IFN- β actively downregulates matrix metalloproteinases (MMPs), stabilizing brain microvascular endothelial tight junctions and preventing vasogenic cerebral edema. Conversely, high-dose **Interferon-Gamma (IFN- γ)** is strictly avoided due to documented hazards of microglial hyper-activation, massive brain swelling, status epilepticus, and permanent epigenetic synaptopathy (chromatin shutdown of synaptic transmission genes) (Fu et al., 2008).

6.5 Targeted Structural Spine and Axonal Regeneration: Neurotrophin-3 (NT-3) and Nerve Growth Factor (NGF)

The seminal findings of Scott et al. (2008) and Jackson (2013) demonstrated that rabies causes severe structural damage to neuronal processes: focal axonal swellings (beading)

and widespread loss of dendritic spines. Clearing the virus halts ongoing insult, but functional neurological recovery requires active structural repair of these injured processes:

- **Neurotrophin-3 (NT-3):** Binds with high selectivity to the TrkC receptor tyrosine kinase, activating the downstream TrkC/ERK/c-Jun signaling pathway (Xu and et al., 2023). NT-3 promotes axonal elongation, stabilizes the neurofilament/microtubule cytoskeleton, reverses rabies-induced axonal beading, and maintains central glial and Schwann cell repair states (Xu and et al., 2023).
- **Nerve Growth Factor (NGF):** Binds to TrkA receptors, driving filamentous actin (F-actin) polymerization, synaptic vesicle clustering, and rapid dendritic spine spinogenesis (Gluska and et al., 2014). Crucially, pathogenic rabies virus specifically hijacks the p75 neurotrophin receptor (p75NTR) to accelerate its retrograde axonal transport machinery (Gluska and et al., 2014); exogenous NGF administration normalizes neurotrophin receptor balance and counteracts transport-associated cytoskeletal distress.

Co-infusion of NT-3 and NGF provides the biochemical foundation to reconstruct degraded dendritic spines and restore axonal continuity during post-clearance recovery.

6.6 Adjunctive Clinical Shields: ICP Control and Anti-Apoptotic Protection

- **3% Hypertonic Saline / Mannitol:** Infused intravenously to maintain serum osmolality at 310–320 mOsm/kg, establishing a hydrostatic osmotic gradient that extracts excess parenchymal free water and controls intracranial pressure (ICP) to protect against herniation.
- **Emricasan:** A synthetic pan-caspase inhibitor that binds caspases 1, 3, 8, and 9, preventing premature apoptosis of infected neurons while ASOs and BCX4430 complete intracellular viral clearance.

7 Component 5: Pediatric Neuroplasticity and Epigenetics

7.1 The Developing Brain and Functional Recovery

Approximately 40% of global rabies cases occur in children under 15 years of age (Liu et al., 2025). Developing brains possess high structural and functional plasticity, characterized by elevated synaptic turnover, dendritic arborization, and cortical remapping potential (James et al., 2026). If viral clearance is achieved before irreversible structural neuronal loss occurs, pediatric neural circuits exhibit substantial capacity to reroute disrupted synaptic pathways.

7.2 Epigenetic Mechanisms and BDNF Expression

Recent pediatric neurotrauma research has elucidated dynamic epigenetic mechanisms of neurorecovery (Heinsberg et al., 2025):

- **BDNF DNA Hypomethylation:** Children recovering from acute brain injury

Therapeutic Agent	Pharmacological Class		Primary Molecular Target	Delivery Platform	
SYN023 (Za-merovimab + Mazorelvimab)	Humanized mAb	IgG1 Cocktail	Non-overlapping RABV-G outer ectodomain epitopes	Stereotactic parenchymal infusion	CED co-infusion
Biparatopic Nanobodies (Rab-E8/H7)	Camelid Domain (~ 15 kDa)	Single-VHH	Cryptic RABV-G epitopes within ~ 20–30 nm synaptic cleft	Stereotactic parenchymal infusion	CED co-infusion
BCX4430 (Galidesivir)	Adenosine Nucleoside analogue	C-Analogue	Viral RdRp RNA chain termination + mTOR autophagy suppression	Stereotactic parenchymal infusion	CED co-infusion
Atorvastatin	HMG-CoA Reductase Inhibitor		Depletion of host lipid droplets; blocks virion budding	Enteral / Systemic maintenance	
Custom Phosphorothioate ASOs / siRNAs	Synthetic Oligonucleotides	Antisense	Targeted cleavage of viral N-gene and L-gene transcripts	Stereotactic parenchymal infusion	CED co-infusion
Interferon-Beta (IFN-β)	Type I Recombinant Cytokine		IFNAR1/2 activation; ISG induction; MMP downregulation	Low-dose CED micro-infusion	titrated
Neurotrophin-3 (NT-3)	Recombinant Neurotrophin		TrkC/ERK pathway; reverses axonal swelling/beading	Stereotactic parenchymal infusion	CED co-infusion
Nerve Growth Factor (NGF)	Recombinant Neurotrophin		TrkA pathway; F-actin polymerization and spino-genesis	Stereotactic parenchymal infusion	CED co-infusion
3% Saline / Mannitol	Hyperosmolar Crystalloid		Serum osmolality (310–320 mOsm/kg); ICP reduction	Intravenous infusion	
Emricasan	Pan-Caspase Inhibitor		Caspases 1, 3, 8, 9; prevents premature somatic apoptosis	Parenchymal / Systemic shield	

Table 4: Comprehensive Pharmacological and Biological Compendium of the RVNA Protocol (Version 2.1).

exhibit significant DNA hypomethylation at the *BDNF* promoter, enhancing transcription of Brain-Derived Neurotrophic Factor (Heinsberg et al., 2025).

- **Rehabilitation-Induced Plasticity:** BDNF upregulation promotes long-term potentiation (LTP), dendritic remodeling, and synaptic repair, providing a validated target for early intensive neurorehabilitation (James et al., 2026; Heinsberg et al., 2025).

7.3 Neuroplastic Recovery Dynamics

Longitudinal neuroimaging following pediatric neurosurgical interventions demonstrates that neural recovery follows a self-repair trajectory, progressing from subcortical structures to cortical networks over a 1-year window (Yan et al., 2025). In the context of RVNA, early neurorehabilitation enhances adaptive neuroplastic changes and reduces long-term functional deficits (James et al., 2026; Yan et al., 2025).

8 Proposed Execution Workflow

The RVNA Protocol is structured into six sequential clinical phases designed for execution in a neuro-intensive care environment:

8.1 Phase 1: Diagnosis & Triage (0–6 hours)

1. Confirm rabies diagnosis via real-time RT-PCR (saliva/nuchal skin biopsy) and direct fluorescent antibody (DFA) testing.
2. Determine elapsed time from initial symptom onset and establish neurological staging.
3. Obtain emergency compassionate-use regulatory and institutional ethical authorization (U.S. Food and Drug Administration, 2025).

8.2 Phase 2: Catheter Placement (2–4 hours)

1. High-resolution 3T MRI stereotactic trajectory planning.
2. Stereotactic placement of bilateral stepped-tip micro-cannulas into thalamic, basal ganglia, and pontine regions.
3. Intraoperative CT/MRI verification of cannula trajectory.

8.3 Phase 3: Regional Hypothermia Induction (1–2 hours)

1. Initiate selective brain cooling to target 32–33 °C.
2. Maintain systemic core temperature at 37 °C using active warming blankets.
3. Establish continuous intracranial pressure (ICP) and brain temperature telemetry.
4. Initiate intravenous 3% hypertonic saline or mannitol titration targeting serum osmolality of 310–320 mOsm/kg.

8.4 Phase 4: Multi-Modal Therapeutic CED Infusion (4–6 hours)

1. Prepare SYN023 (0.3 mg/kg equivalent parenchymal volume) and multi-epitope mAbs (sites II, III, IV) (Liu et al., 2025; Ren et al., 2025).
2. Prepare anti-rabies biparatopic VHH cocktail (50–100 μ g/kg) (Terry et al., 2016).
3. Co-formulate BCX4430 (50–100 μ M equivalent parenchymal target) and custom N-/L-gene ASOs (Xie et al., 2023).
4. Add trace gadoteridol (1.5 mM) for live intraoperative distribution surveillance.
5. Initiate CED infusion at 0.5 μ L/min, ramping to 1.5–2.5 μ L/min under continuous T1-weighted MRI monitoring (Narsinh et al., 2025; Munjal et al., 2025).
6. Administer oral/enteral atorvastatin to suppress host lipid droplet-mediated viral budding (Zhao et al., 2022).

8.5 Phase 5: Monitoring, Neurotrophic Repair & Support (3–5 days)

1. Controlled rewarming at 0.25 $^{\circ}$ C/hr.
2. Continuous neuro-ICU supportive care and continuous EEG monitoring for subclinical seizures.
3. Co-infuse Neurotrophin-3 (NT-3) and Nerve Growth Factor (NGF) via CED to drive dendritic spine spinogenesis and reverse axonal beading (Scott et al., 2008; Xu et al., 2023; Gluska et al., 2014).
4. Titrate low-dose IFN- β micro-infusion for endothelial microvascular stabilization.
5. Serial CSF viral load and biomarker quantification.
6. Maintain cerebral perfusion pressure (CPP > 65 mmHg).

8.6 Phase 6: Neurorehabilitation & Epigenetic BDNF Mobilization (Ongoing)

1. Initiate early multisensory and motor stimulation protocols.
2. Longitudinal cognitive, speech, and physical rehabilitation (James et al., 2026).
3. Monitor neuroplastic recovery and BDNF epigenetic hypomethylation profiles (Heinsberg et al., 2025).

9 Ethical Justification and Regulatory Framework

9.1 Compassionate Use Framework

Expanded access (compassionate use) provides an established regulatory pathway for patients with life-threatening conditions to receive investigational therapeutics outside of clinical trials when **no comparable or satisfactory alternative therapies exist** (U.S. Food and Drug Administration, 2025).

The RVNA Protocol is formulated specifically for this population:

- No established curative therapy exists for symptomatic rabies encephalomyelitis (Jackson, 2011).
- Current standard of care is restricted to terminal palliative sedation (Lacy and Jackson, 2024).
- Untreated disease carries a virtually 100% mortality rate.

9.2 FDA Expanded Access Criteria

Under FDA Expanded Access regulations (21 CFR Part 312.300 / Emergency IND pathways), experimental therapies may be authorized when the potential benefit justifies potential risks (U.S. Food and Drug Administration, 2025):

“Investigational medical products have not yet been approved or cleared by FDA and FDA has not found these products to be safe and effective for their specific use. Furthermore, the investigational medical product may, or may not, be effective in the treatment of the condition, and use of the product may cause unexpected serious side effects.” (U.S. Food and Drug Administration, 2025)

In the absence of existing alternatives, the catastrophic natural history of rabies encephalomyelitis justifies consideration under expanded access.

9.3 Pediatric Ethical Considerations

Because approximately 40% of global rabies deaths occur in children, the ethical imperative to provide active therapeutic options is paramount. Pediatric patients possess the greatest potential for neurodevelopmental functional recovery following acute insults; providing an active clearance pathway respects the patient’s right to life-saving intervention over passive palliation.

9.4 Informed Consent

Informed consent protocols must explicitly delineate:

1. The experimental nature and unproven clinical efficacy of the multi-modal stack;
2. Known neurosurgical risks (intracranial hemorrhage, infection, edema);
3. Alternative options (palliative comfort measures);
4. The unconditional right of legal representatives to withdraw from the protocol at any stage; and
5. Comprehensive data collection for international scientific learning.

10 Limitations & Future Work

10.1 Limitations

Table 5: Key limitations of the RVNA Protocol and proposed mitigations.

Limitation	Description	Mitigation
Untested Combination	The complete multi-modal protocol has not been evaluated in combined in vivo trials.	Prioritize staged rodent and non-human primate symptomatic challenge studies.
Nanobody & ASO Supply	Anti-rabies nanobodies (VHH) and ASOs require dedicated GMP synthesis batches.	Establish standardized <i>Pichia pastoris</i> biomanufacturing and oligonucleotide synthesis pipelines (Terryn et al., 2016).
Neurosurgical Risks	Catheter insertion carries risks of intracranial hemorrhage, reflux, and infection.	Utilize stepped-design cannulas, micro-infusion rates ($< 3.0 \mu\text{L}/\text{min}$), and real-time MRI guidance (Parvar et al., 2025; Narsinh et al., 2025).
Intracranial Pressure (ICP)	Parenchymal infusion into inflamed tissue risks elevating ICP and causing edema.	Continuous ICP telemetry, osmoprotective agents (mannitol / 3% hypertonic saline), and ramped flow protocols.
Structural Spine Loss	Rabies virus induces axonal beading and dendritic spine collapse via F-actin depolymerization.	Co-infusion of neurotrophic factors (NT-3, NGF) and early pediatric neurorehabilitation (Scott et al., 2008; Xu and et al., 2023; Gluska and et al., 2014).
Animal Model Gap	Lack of established large-animal models replicating symptomatic CNS viral clearance.	Implement systematic feline or canine translational models alongside compassionate clinical registry data.
Viral Load Quantification	Real-time measurement of parenchymal viral load and synaptic saturation is challenging.	Develop serial CSF biomarker panels and microdialysis sampling protocols.
Thermal Gradient Control	Selective brain cooling requires precise temperature titration without systemic shivering.	Closed-loop nasopharyngeal / endovascular heat exchangers with active core warming blankets (37°C).
Post-Lysis Inflammation	Rapid extracellular virion neutralization may trigger local neuroinflammatory cascades.	Adjunctive neuroprotective modulation, low-dose IFN- β endothelial stabilization, and hypothermia.

10.2 Future Work

1. **Preclinical In Vivo Validation:** Evaluate combined RVNA components (CED + regional hypothermia + multi-epitope mAbs + biparatopic VHH + BCX4430 + NT-3/NGF) in symptomatic animal challenge models.
2. **Nanobody Optimization:** Engineer multivalent and biparatopic nanobodies targeting conserved viral glycoprotein epitopes (Terry et al., 2016).
3. **Computational Fluid Dynamics (CFD) Modeling:** Model CED distribution dynamics in infected cerebral tissue to optimize stereotactic trajectory planning (Parvar et al., 2025).
4. **Compassionate Use Clinical Registry:** Establish an international multi-center registry to document outcomes of individual compassionate-use cases (U.S. Food and Drug Administration, 2025).
5. **Formulation Optimization:** Develop specialized nanocarrier and hydrogel formulations for stabilized intracranial CED antibody and neurotrophin delivery (Parvar et al., 2025).

11 Roadmap to RVNA Protocol Version 3.0: Open-Source CED Simulation Engine & Transport Equations

To accelerate global clinical readiness and stereotactic precision, the forthcoming **RVNA Protocol Version 3.0** will release a dedicated, fully open-source finite-element computational fluid dynamics (CFD) simulation suite. This platform solves the coupled biophysical equations of fluid momentum, multi-species convection-dispersion mass transfer, and tissue poroelasticity across patient-specific neuroanatomical meshes.

11.1 Governing Biophysical Transport Equations

11.1.1 1. Interstitial Darcy Fluid Flow in Anisotropic Cerebral Parenchyma

Convective fluid velocity $\mathbf{v}$ generated by positive micro-infusion hydrostatic pressure gradients in porous brain tissue is governed by Darcy’s Law:

$$\mathbf{v} = -\mathbf{K}(\mathbf{x})\nabla p \quad (2)$$

where p is interstitial fluid pressure (Pa) and $\mathbf{K}(\mathbf{x}) = \frac{\mathbf{k}(\mathbf{x})}{\mu}$ is the hydraulic conductivity tensor ($\text{m}^2/(\text{Pa} \cdot \text{s})$). To account for preferential bulk flow along white matter tracts, $\mathbf{k}(\mathbf{x})$ is constructed directly from patient-specific Diffusion Tensor Imaging (DTI) fractional anisotropy matrices:

$$\mathbf{k}(\mathbf{x}) = k_{\text{iso}} \left[(1 - \alpha_{\text{ani}})\mathbf{I} + \alpha_{\text{ani}}\mathbf{v}_1\mathbf{v}_1^T \right] \quad (3)$$

where $\mathbf{v}_1$ is the principal DTI tractography eigenvector and $\alpha_{\text{ani}} \in [0, 1]$ is the local tract anisotropy coefficient.

11.1.2 2. Fluid Mass Conservation and Microvascular Sinks

Steady-state conservation of fluid mass within deformable porous brain tissue satisfies:

$$\nabla \cdot \mathbf{v} = S_{\text{infusion}} - L_p (p - p_{\text{microvasc}}) \quad (4)$$

where $S_{\text{infusion}}(\mathbf{x}, t)$ is the cannula volumetric infusion rate (0.1–5.0 $\mu\text{L}/\text{min}$), L_p is the microvascular filtration coefficient, and $p_{\text{microvasc}}$ is capillary venous pressure.

11.1.3 3. Multi-Species Convection-Dispersion Mass Transfer

For each therapeutic biologic and small molecule in the RVNA stack (SYN023 full-length IgG, biparatopic VHH nanobodies, BCX4430, custom ASOs, NT-3, and NGF), spatial and temporal concentration distributions $C_i(\mathbf{x}, t)$ are governed by the convection-dispersion-reaction transport equation:

$$\frac{\partial(\theta C_i)}{\partial t} + \nabla \cdot (\mathbf{v} C_i) = \nabla \cdot (\mathbf{D}_i^* \nabla C_i) - k_{\text{elim},i} C_i - k_{\text{on},i} C_i (B_{\text{max},i} - B_i) + k_{\text{off},i} B_i \quad (5)$$

where:

- θ is the extracellular volume fraction ($\theta \approx 0.20$);
- $\mathbf{D}_i^* = \frac{D_{0,i}}{\lambda_{\text{tort}}^2} \mathbf{I} + \mathbf{D}_{\text{mech}}(\mathbf{v})$ is the effective hydrodynamic dispersion tensor incorporating tissue tortuosity ($\lambda_{\text{tort}} \approx 1.6$) and velocity-dependent mechanical dispersion;
- $k_{\text{elim},i}$ is the first-order clearance and enzymatic degradation rate;
- B_i is the concentration of drug-target complexes (e.g., antibody-virion or neurotrophin-receptor conjugates), with association rate $k_{\text{on},i}$ and dissociation rate $k_{\text{off},i}$.

11.1.4 4. Poroelastic Tissue Deformation and Reflux Prevention

Infusion-induced tissue displacement $\mathbf{u}$ and backflow risk along the catheter tract are modeled using Biot’s poroelasticity formulation:

$$\nabla \cdot \boldsymbol{\sigma}' - \alpha \nabla p = \mathbf{0}, \quad \boldsymbol{\sigma}' = 2G\boldsymbol{\varepsilon} + \lambda \text{tr}(\boldsymbol{\varepsilon}) \mathbf{I} \quad (6)$$

where $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$ is the strain tensor, $\alpha \approx 1$ is the Biot coefficient, and G, λ are the Lamé elasticity moduli of brain tissue.

Version 3.0 Open-Source Release Milestones

The upcoming Version 3.0 release will provide:

1. **FEniCS / OpenFOAM Simulation Engine:** Modular open-source finite element code solving equations (3)–(7) for multi-drug CED co-infusion.
2. **Automated V_d/V_i Predictive Optimization:** Algorithmic catheter trajectory optimization computing distribution-to-infusion volume ratios ($V_d/V_i = 2.0$ – 5.0) across bilateral thalamic and brainstem nuclei.
3. **Clinical DICOM/DTI Importer:** Automated pipeline transforming patient-specific MRI DICOM datasets into high-density 3D volumetric finite-element meshes.

12 Conclusion

“Although preventative therapy for rabies is highly successful after recognized exposures, there is no established therapy that is effective for patients who develop rabies encephalomyelitis, which remains an important therapeutic challenge for the future.” (Jackson, 2011)

For decades, the clinical management of symptomatic rabies has remained trapped between therapeutic despair and unscientific coma protocols. The RVNA Protocol proposes a fundamental paradigm shift: replacing passive metabolic sedation with a targeted, biophysically grounded, active clearance architecture.

By combining Convection-Enhanced Delivery, dual full-IgG and synaptic-penetrating nanobodies, targeted regional brain hypothermia, and pediatric neuroplastic potential, the RVNA framework provides an actionable blueprint for translational investigation and compassionate clinical trial in medicine’s most lethal infectious challenge.

“More basic research is needed to improve our understanding of the mechanisms involved in rabies pathogenesis, which will hopefully facilitate the development of novel therapeutic approaches in the future.” (Jackson, 2011)

The RVNA Protocol is a contribution to that future.

Acknowledgements

The authors thank the global community of neurovirologists, neurosurgeons, and biomedical engineers whose open scientific contributions in rabies pathogenesis, convection-enhanced delivery, antibody engineering, and therapeutic hypothermia informed this systems architecture. No external funding was received for this work.

Medical and Legal Disclaimer

This document describes a theoretical biomedical systems engineering proposal for scientific research and expanded-access evaluation. It does not constitute formal medical advice. All clinical procedures, expanded-access applications, and surgical interventions must be evaluated and administered by licensed medical specialists in accordance with applicable national laws, institutional review boards (IRBs), and regulatory health authorities.

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